Unit IV – CPU Scheduling and Algorithm

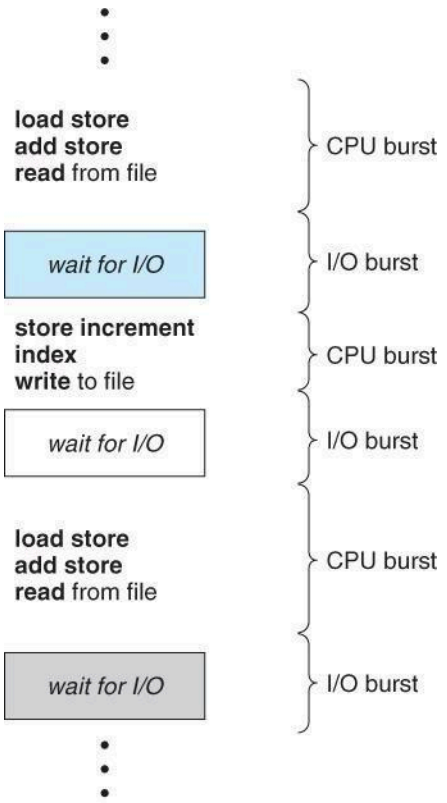
Section 4.1 Scheduling types

Scheduling Objectives

- Be Fair while allocating resources to the processes
- Maximize throughput of the system
- Maximize number of users receiving acceptable response times.
- Be predictable
- Balance resource use
- Avoid indefinite postponement
- Enforce Priorities
- Give preference to processes holding key resources
- Give better service to processes that have desirable behaviour patterns

CPU and I/O Burst Cycle:

- Process execution consists of a cycle of CPU execution and I/O wait.
- Processes alternate between these two states.
- Process execution begins with a CPU burst, followed by an I/O burst, then another CPU burst ... etc
- The last CPU burst will end with a system request to terminate execution rather than with another I/O burst.
- The duration of these CPU burst have been measured.
- An I/O-bound program would typically have many short CPU bursts, A CPU-bound program might have a few very long CPU bursts.
- This can help to select an appropriate CPU-scheduling algorithm.



Preemptive Scheduling:

- Preemptive scheduling is used when a process switches from running state to ready state or from waiting state to ready state.
- The resources (mainly CPU cycles) are allocated to the process for the limited amount of time and then is taken away, and the process is again placed back in the ready queue if that process still has CPU burst time remaining.
- That process stays in ready queue till it gets next chance to execute.

Non-Preemptive Scheduling:

- Non-preemptive Scheduling is used when a process terminates, or a process switches from running to waiting state.
- In this scheduling, once the resources (CPU cycles) is allocated to a process, the process holds the CPU till it gets terminated or it reaches a waiting state.
- In case of non-preemptive scheduling does not interrupt a process running CPU in middle of the execution.
- Instead, it waits till the process complete its CPU burst time and then it can allocate the CPU to another process.

Basis for Comparison	Preemptive Scheduling	Non Preemptive Scheduling
Basic	The resources are allocated to a process for a limited time.	Once resources are allocated to a process, the process holds it till it completes its burst time or switches to waiting state.
Interrupt	Process can be interrupted in between.	Process can not be interrupted till it terminates or switches to waiting state.
Starvation	If a high priority process frequently arrives in the ready queue, low priority process may starve.	If a process with long burst time is running CPU, then another process with less CPU burst time may starve.
Overhead	Preemptive scheduling has overheads of scheduling the processes.	Non-preemptive scheduling does not have overheads.
Flexibility	Preemptive scheduling is flexible.	Non-preemptive scheduling is rigid.
Cost	Preemptive scheduling is cost associated.	Non-preemptive scheduling is not cost associative.

Scheduling Criteria

- There are several different criteria to consider when trying to select the "best" scheduling algorithm for a particular situation and environment, including:
    - **CPU utilization** - Ideally the CPU would be busy 100% of the time, so as to waste 0 CPU cycles. On a real system CPU usage should range from 40% (lightly loaded) to 90% (heavily loaded.)
    - **Throughput** - Number of processes completed per unit time. May range from 10 / second to 1 / hour depending on the specific processes.

- **Turnaround time** - Time required for a particular process to complete, from submission time to completion.
- **Waiting time** - How much time processes spend in the ready queue waiting their turn to get on the CPU.
- **Response time** - The time taken in an interactive program from the issuance of a command to the ***commence*** of a response to that command.

In brief:

Arrival Time: Time at which the process arrives in the ready queue. **Completion Time:** Time at which process completes its execution. **Burst Time:** Time required by a process for CPU execution. **Turn Around Time:** Time Difference between completion time and arrival time. Turn Around Time = Completion Time – Arrival Time

Waiting Time(W.T): Time Difference between turnaround time and burst time. Waiting Time = Turn Around Time – Burst Time

4.2 Types of Scheduling Algorithm

(a) First Come First Serve (FCFS)

In FCFS Scheduling

- The process which arrives first in the ready queue is firstly assigned the CPU.
- In case of a tie, process with smaller process id is executed first.
- It is always non-preemptive in nature.
- Jobs are executed on first come, first serve basis.
- It is a non-preemptive, pre-emptive scheduling algorithm.
- Easy to understand and implement.
- Its implementation is based on FIFO queue.
- Poor in performance as average wait time is high.

Advantages-

- It is simple and easy to understand.
- It can be easily implemented using queue data structure.
- It does not lead to starvation.

Disadvantages-

- It does not consider the priority or burst time of the processes.
- It suffers from convoy effect i.e. processes with higher burst time arrived before the processes with smaller burst time.

Example 1:

Q. Consider the following processes with burst time (CPU Execution time). Calculate the average waiting time and average turnaround time?

Process id	Arrival time	Burst time/CPU execution time
P1	0	2
P2	1	3
P3	2	5
P4	3	4
P5	4	6

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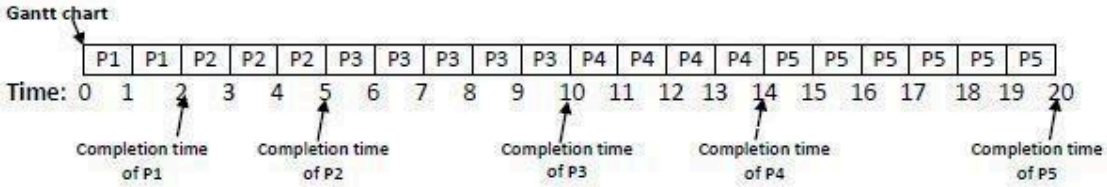
Gantt chart



Turnaround time= Completion time – Arrival time

Waiting time= Turnaround time – Burst time

Process id	Arrival time	Burst time	Completion time	Turnaround time	Waiting time
P1	0	2	2	2-0=2	2-2=0
P2	1	3	5	5-1=4	4-3=1
P3	2	5	10	10-2=8	8-5=3
P4	3	4	14	14-3=11	11-4=7
P5	4	6	20	20-4=16	16-6=10

Average turnaround time= $\sum_{i=0}^{n}$ Turnaround time(i)/n where, n= no. of process

Average waiting time= $\sum_{i=0}^{n}$ Wating time(i)/n where, n= no. of process

Average turnaround time= 2+4+8+11+16/5 =41/5 =8.2

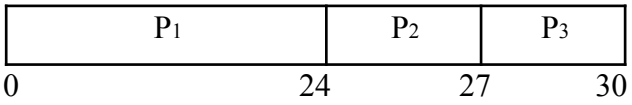
Average waiting time= 0+1+3+7+10/5 = 21/5 =4.2

Example 2:

Consider the processes P1, P2, P3 given in the below table, arrives for execution in the same order, with Arrival Time 0, and given Burst Time,

PROCESS	ARRIVAL TIME	BURST TIME
P1	0	24
P2	0	3
P3	0	3

Gantt chart



PROCESS	WAIT TIME	TURN AROUND TIME
P1	0	24
P2	24	27
P3	27	30

Total Wait Time = 0 + 24 + 27 = 51 ms

Average Waiting Time = (Total Wait Time) / (Total number of processes) = 51/3 = 17 ms

Total Turn Around Time: 24 + 27 + 30 = 81 ms

Average Turn Around time = (Total Turn Around Time) / (Total number of processes)
= 81 / 3 = 27 ms

Throughput = 3 jobs/30 sec = 0.1 jobs/sec

Example 3:

Consider the processes P1, P2, P3, P4 given in the below table, arrives for execution in the same order, with given Arrival Time and Burst Time.

PROCESS	ARRIVAL TIME	BURST TIME
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Gantt chart

P_1	P_2	P_3	P_4

0 8 12 21 26

PROCESS	WAIT TIME	TURN AROUND TIME
P1	0	8 – 0 = 8
P2	8 – 1 = 7	12 – 1 = 11
P3	12 – 2 = 10	21 – 2 = 19
P4	21 – 3 = 18	26 – 3 = 23

Total Wait Time:= 0 + 7 + 10 + 18 = 35 ms

Average Waiting Time = (Total Wait Time) / (Total number of processes)= 35/4 = 8.75 ms

Total Turn Around Time: 8 + 11 + 19 + 23 = 61 ms

Average Turn Around time = (Total Turn Around Time) / (Total number of processes)
61/4 = 15.25 ms

Throughput: 4 jobs/26 sec = 0.15385 jobs/sec

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